



ME 142- ENGINEERING DRAWING & GRAPHICS

HITEC UNIVERSITY TAXILA
Department of Mechanical Engineering

A vertical image on the left side of the slide shows a hand with the index finger pointing at a technical drawing or blueprint. The drawing contains various lines, circles, and text, typical of engineering or architectural plans.

Units of Length

- Miles, yard, feet and inches
- Km, meter, cm ,mm and μm
- Conversion of one unit to other
- Unit prefixes
- Area?
- Volume?

A vertical image on the left side of the slide shows a hand with a pointing finger directed at a technical drawing or blueprint. The drawing contains various lines, circles, and text, typical of engineering or architectural plans.

Ratio

- A:B , 2:3
- $\frac{1}{2}$, $\frac{7}{9}$
- x is five times greater than y
- Ratio between a and b is 3 ratio 1
- y is half of x



Recommended Books

Text Books

- Engineering Drawing with an introduction to AutoCAD
By: *Dhananjay A Jolhe(Text Book)*
- Engineering Drawing and Graphics by N.D. Bhatt

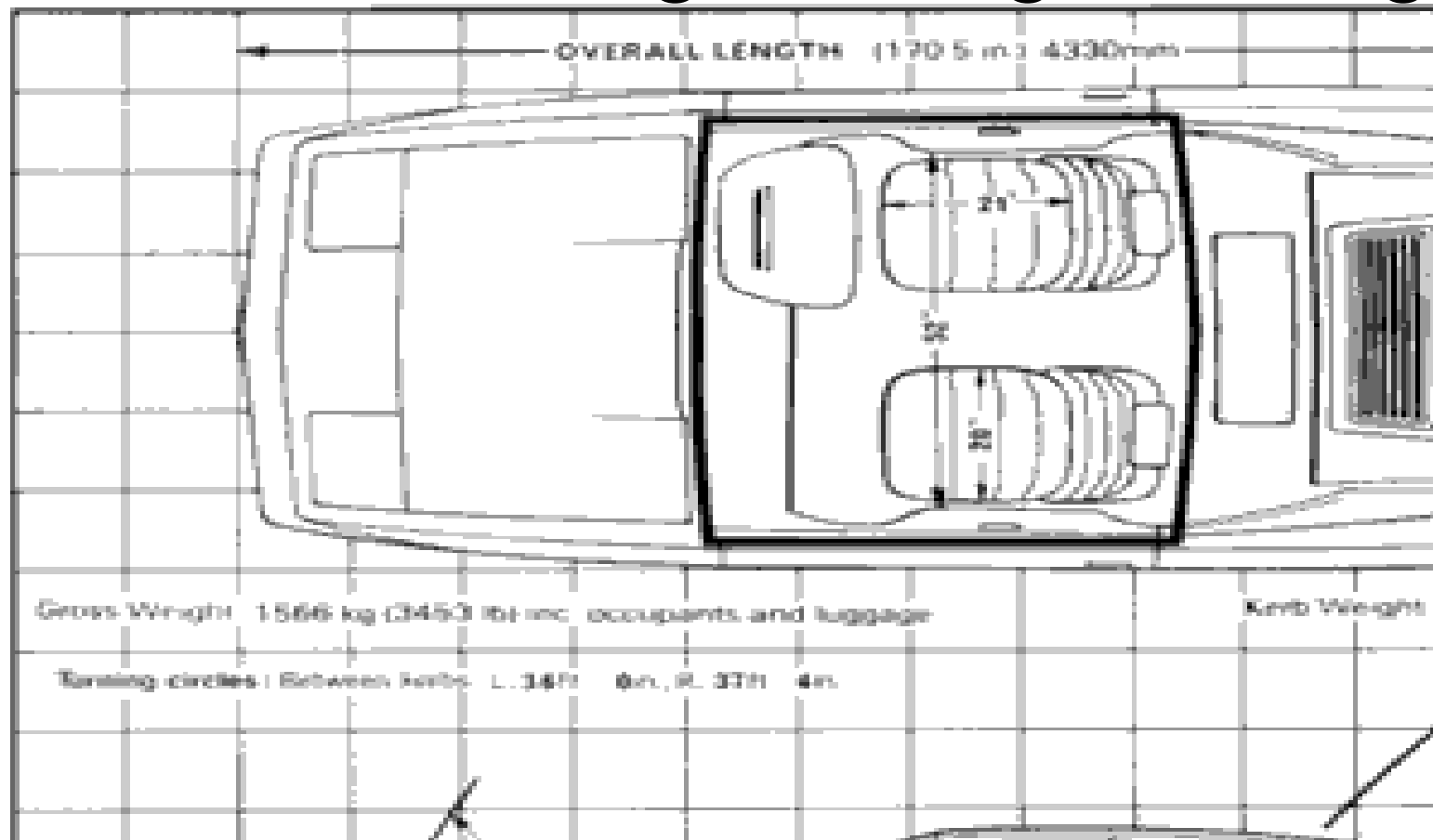
Engineering Drawing

- What is a drawing?



Engineering Drawing

- What is an engineering drawing?



A vertical image on the left side of the slide shows a hand pointing with the index finger at a technical drawing on a piece of paper. The drawing contains various lines, circles, and text, typical of an engineering blueprint.

Why Engineering Drawing?

It is a language used in the design process

– It is used to:

- Visualize
- Communicate
- Document

the engineering design

A vertical image on the left side of the slide shows a person's hand pointing with their index finger at a technical drawing. The drawing is on a light blue background and contains various lines, circles, and text, typical of an engineering blueprint.

An Engineering Drawing

- An **engineering drawing** is a type of ***technical drawing***, used to fully and clearly define requirements for engineered items, and is usually created in accordance with standardized conventions for layout, *nomenclature, interpretation, appearance, size*, etc.
- Its purpose is to accurately and unambiguously capture all the geometric features of a product or a component.
- The end goal of an engineering drawing is to convey all the required information that will allow a manufacturer to produce that component.

A vertical image on the left side of the slide shows a hand with the index finger pointing down at a technical drawing. The drawing is on a light blue background and contains various lines and text, including the words 'FOLLOWING' and 'DRAWING'.

Types of Engineering Drawings

- Free Hand Sketches
- Technical Drawing
 - Manual Drawing
 - Computer Aided Drawing/Drafting

A vertical image on the left side of the slide shows a hand with the index finger pointing at a technical drawing. The drawing is a blue-lined sketch on white paper, showing mechanical components and dimensions.

Technical Drawings

- Three main types of technical drawing
 - Detail Drawings (Shop drawings)
 - Assembly & Sub-Assembly Drawings
 - Exploded View Drawings

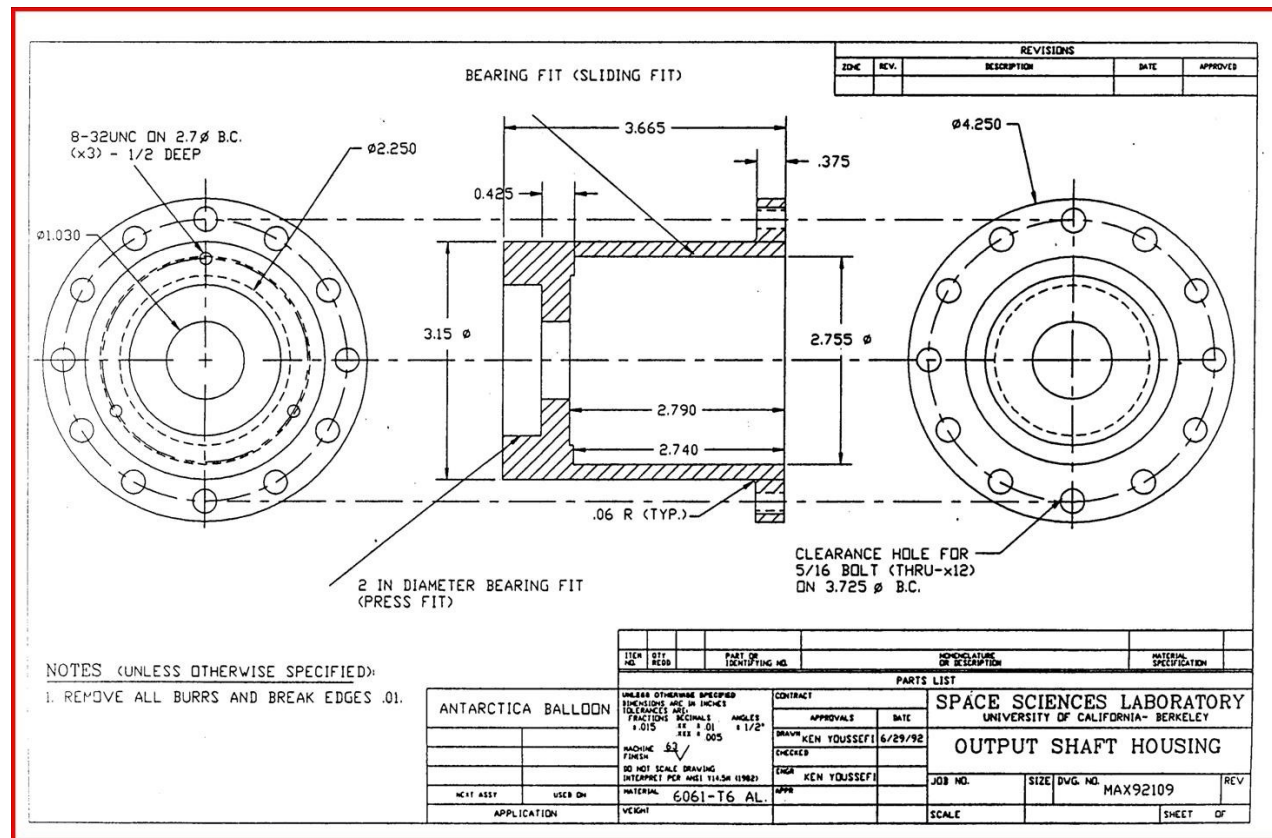
A vertical strip on the left side of the slide shows a close-up of a person's hand. The index finger is pointing down towards a technical drawing on a piece of paper. The drawing contains various lines, circles, and some handwritten text, typical of engineering sketches.

Engineering Drawing

- Sketching??
- Free Hand drawings
- Drafting??
- Technical Drawings

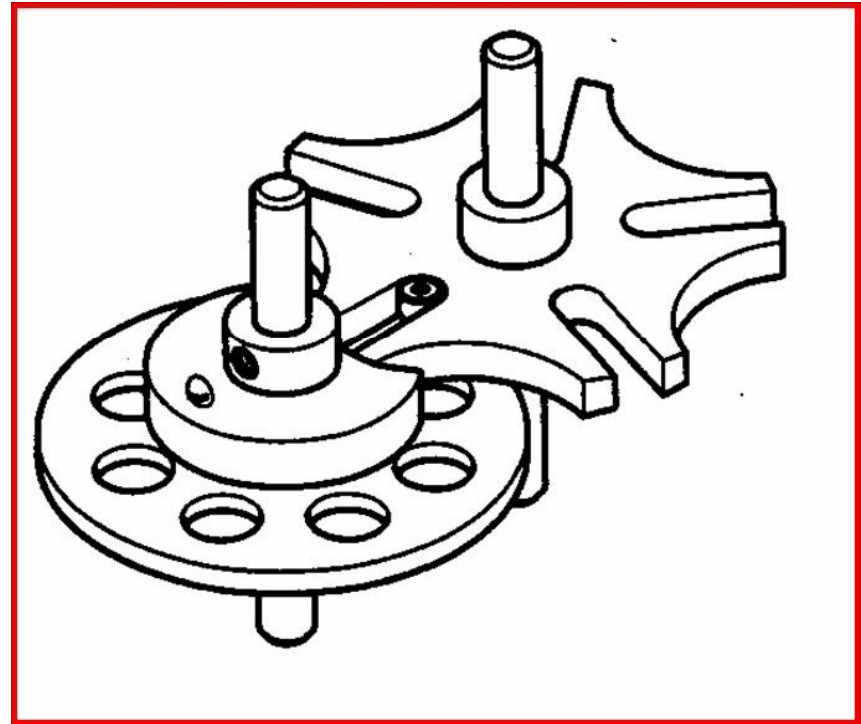
Detail Drawings

- Two dimensional multi-view drawings of each component for the purpose of fabrication.



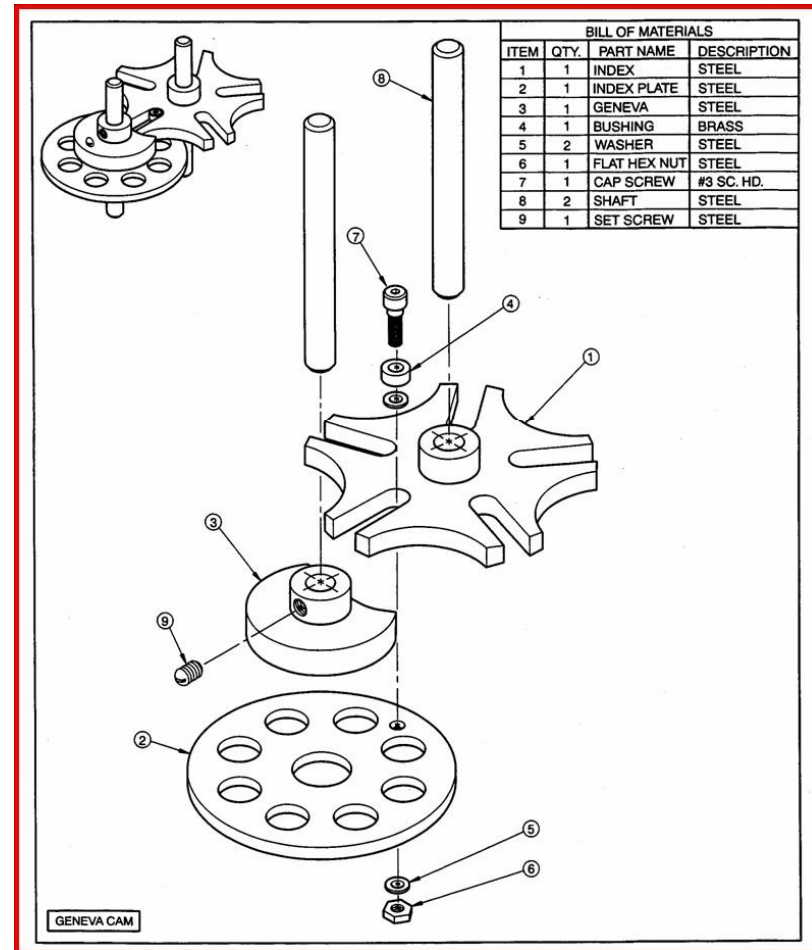
Assembly Drawings

- Three dimensional drawings show all parts assembled. Parts list . normally no dimensions are given in the assembly drawings.



Exploded View Drawings

- Three dimensional drawings showing how parts should be assembled.





Common Features

- **Geometry**
 - the shape of the object represented as views.
 - how the object will look when it is viewed from various standard directions.
- **Dimensions**
 - the size of the object is captured in accepted units.
- **Tolerances**
 - the allowable variations for each dimension.
- **Material –**
 - represents what the item is made of.
- **Finish –**
 - specifies the surface quality of the item, functional or cosmetic.



Drawing Instruments

List of Items commonly used in Drafting

1. Drawing Board
2. T-Square & Drafting Machine
3. Drawing Instrument Box
4. Triangles
5. Scales (Ruler)
6. Protractor
7. Templates
8. Adjustable Curves



Drawing Instruments (contd.)

- 9. Drafting Pencils
- 10. Eraser & Eraser Shield
- 11. Drawing Sheets
- 12. Adhesive Tape
- 13. Dusting Brush
- 14. Calculator

Drawing Pencils

Hard Lead: Sharp but grey lines

Soft Lead: Black but fuzzy lines

6H – For light layout work with accuracy

4H – Dimension lines, Center line and Dashed lines

2H – For visible object lines

H – For lettering and free hand



Drawing Pencil Grades



Softest

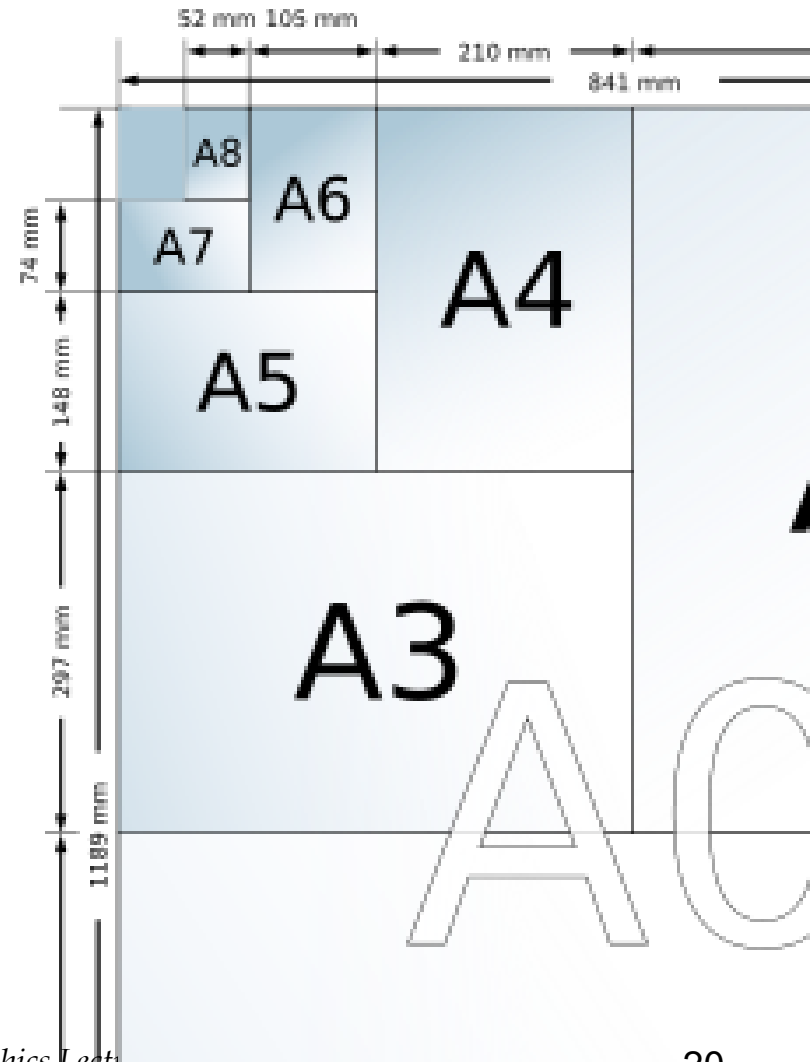
SOFT					MEDIUM					HARD						
6B	5B	4B	3B	2B	B	HB	F	H	2H	3H	4H	5H	6H	7H	8H	9H

Hardest

Drawing Paper

•**Metric System:** Metric drawing sizes are based on the A0 size, having an area of 1 m² and a length to width ratio of 1:√ 2.

•**Inch System:** Drawing sizes in the inch system are based on dimensions of commercial letterheads, 8.5x11in.

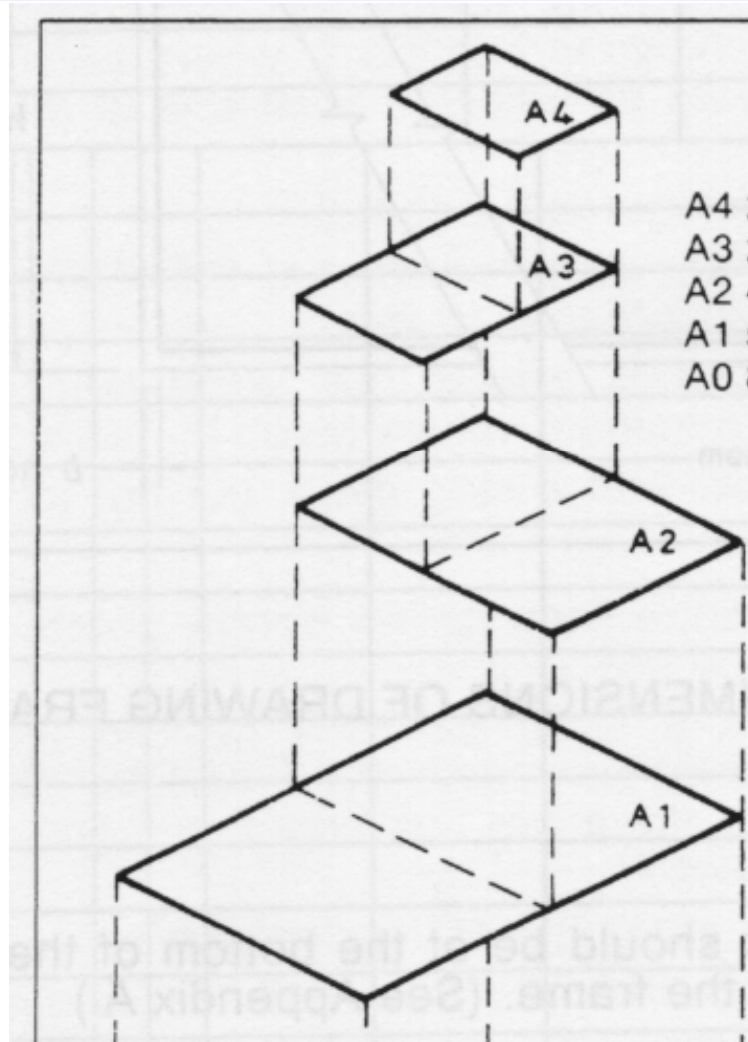


Drawing Paper Sizes

METRIC DRAWING SIZES (mm)		
Drawing Sizes	Border Size	Overall Paper Size
A4	190X267	210 X 297
A3	277 X 390	297 X 420
A2	400 X 564	420 X 594
A1	574 X 811	594 X 841
A0	821 X 1159	841 X 1189

INCH DRAWING SIZES (in.)		
Drawing Sizes	Border Size	Overall Paper Size
A	8.00 X 10.50	8.50 X 11.00
B	10.50 X 16.50	11.00 X 17.00
C	16.25 X 21.25	17.00 X 22.00
D	21.00 X 33.00	22.00 X 34.00
E	33.00 X 43.00	34.00 X 44.00

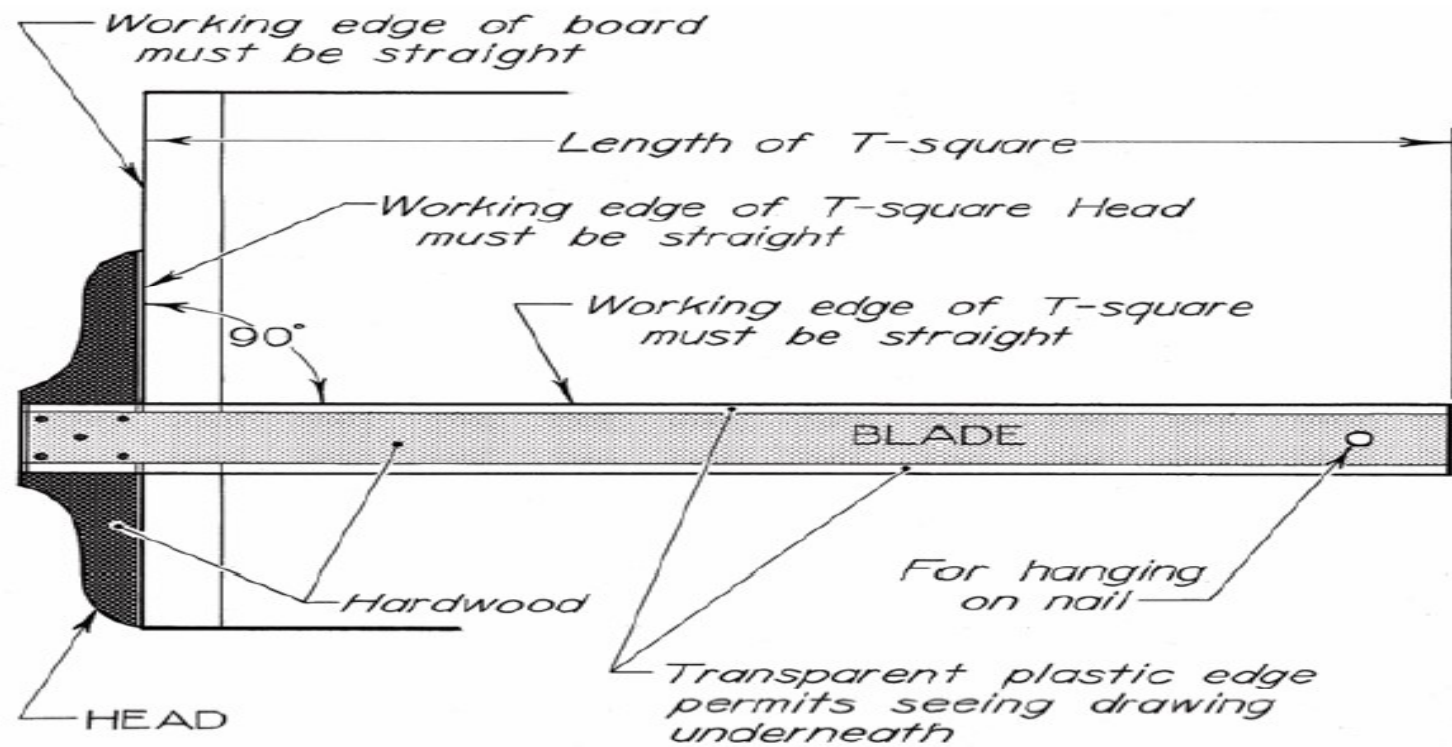
Relationship of A sizes



Drawing Board



T-Square



T-Square

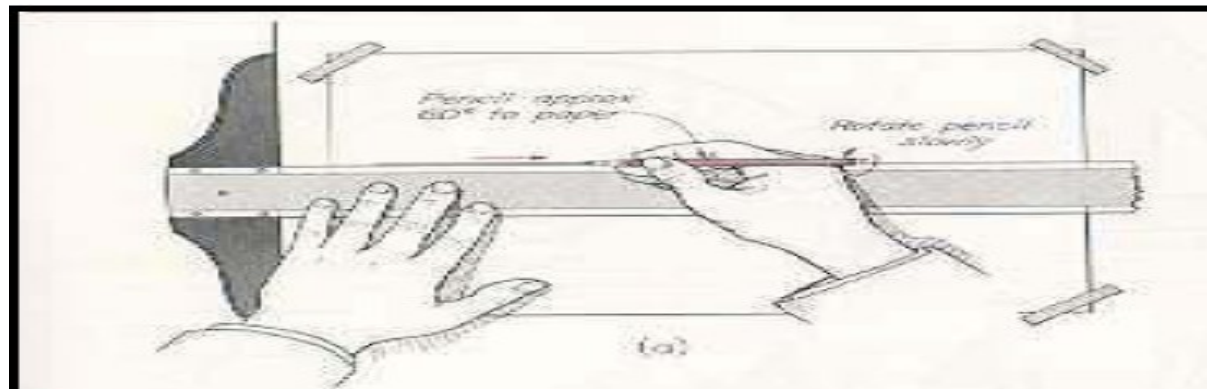


Fig. 2.20 Drawing a Horizontal Line.

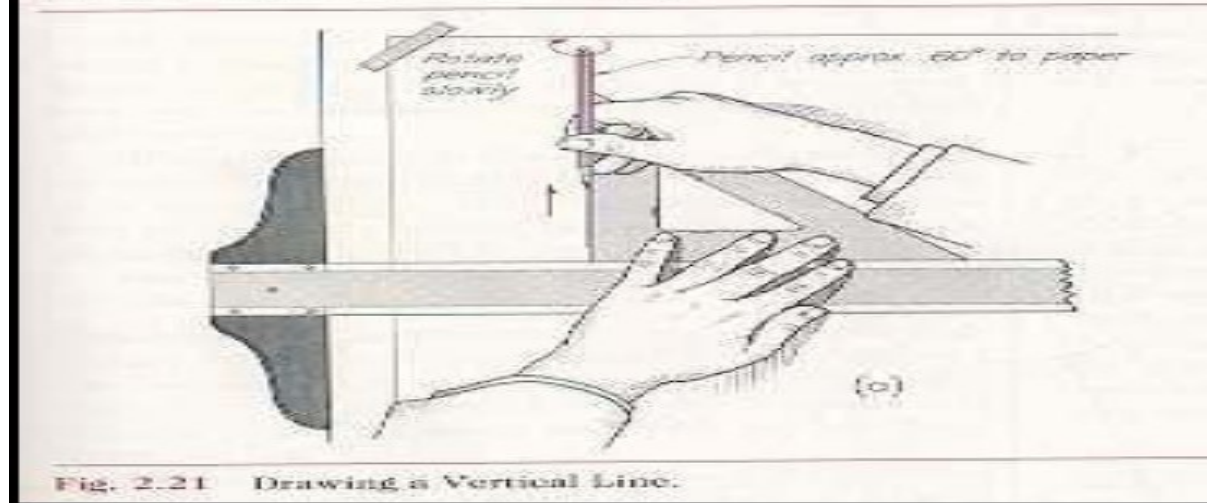
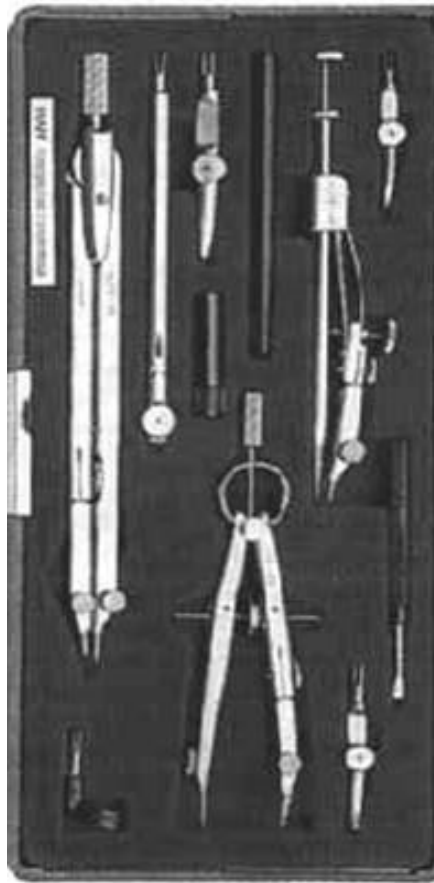


Fig. 2.21 Drawing a Vertical Line.

Drafting Machine



Instrument Box



Friction Compass



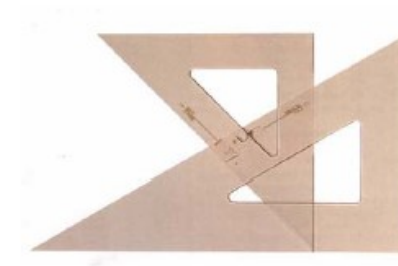
Bow Compass



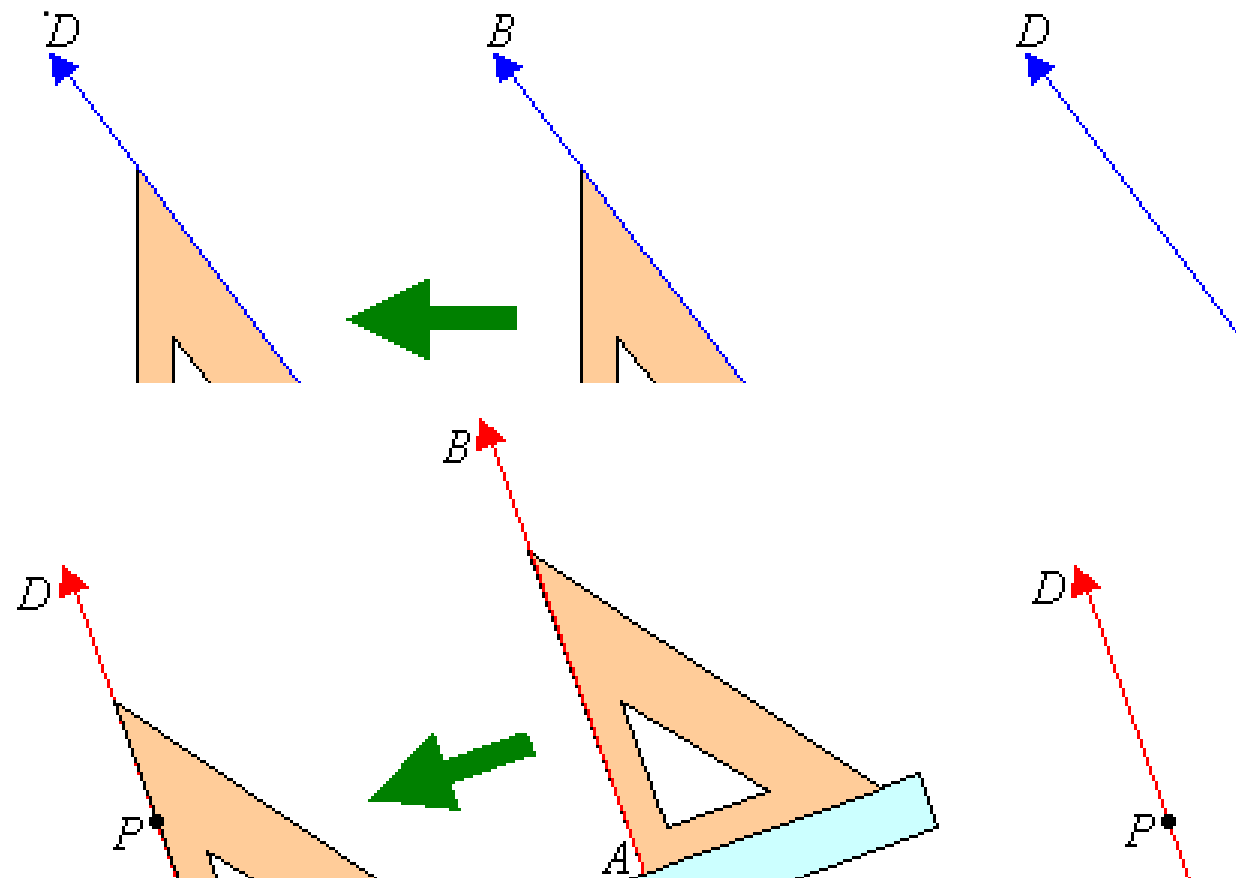
Divider



Triangles (Set Squares)

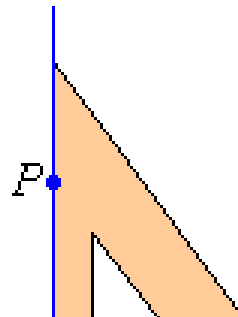
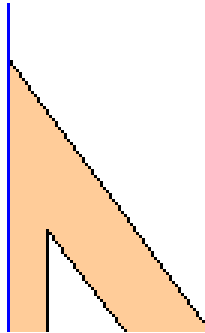


Parallel Lines

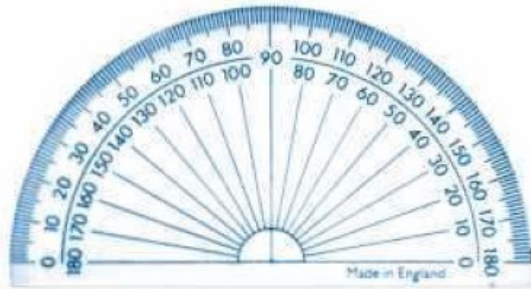


and $PQ \parallel RS$.

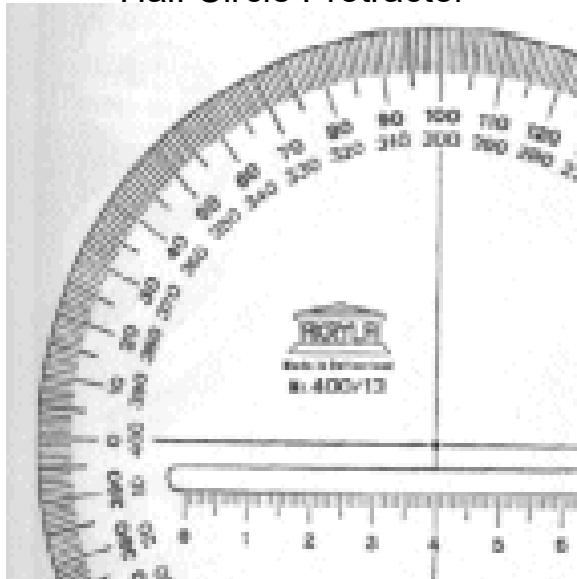
Perpendicular Lines



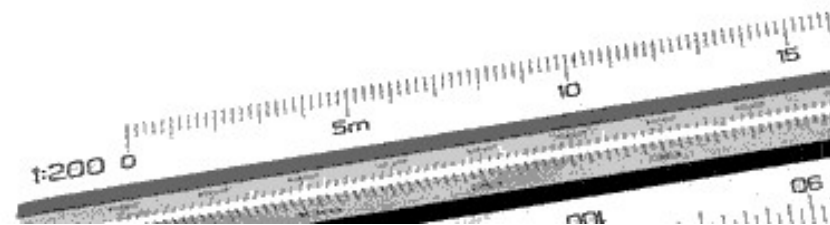
Protractors and Scales



Half Circle Protractor

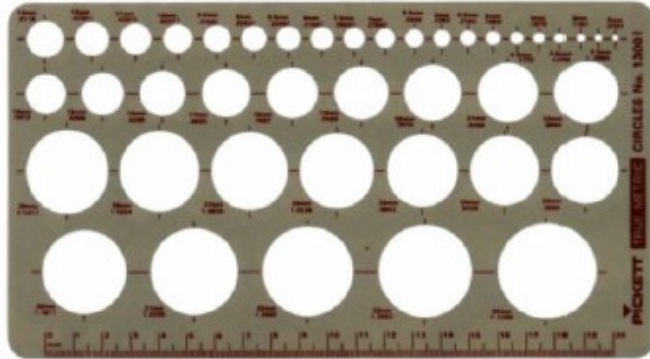


Full Circle Protractor

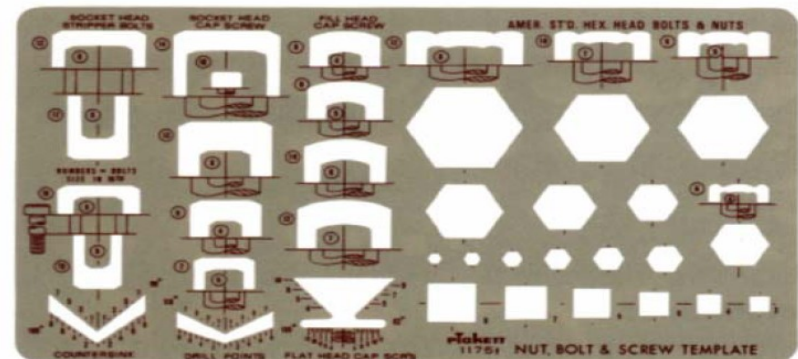


Scales

Templates



Circle Template



Nut, Bolt and Screw Template

Adjustable Curves



French Curves



Flex Curves